

Project Proposal: Predicting Public Passenger Vehicle Inspection Outcomes in Chicago

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1. Dataset Selection **Dataset Name:** Public Passenger Vehicle Inspection Schedule **Source:** data.gov (City of Chicago Data Portal) **Description:** This dataset contains approximately 35,000 records detailing the inspection schedules and results for public passenger vehicles (taxis, ambulances, liveries, etc.) licensed by the City of Chicago. **Key Features:** The dataset includes meaningful columns, such as Vehicle Type, Make, Model, Year, Company Name, Inspection Type (e.g., Scheduled, Retest), Vehicle Status, and the target variable, Result (e.g., Passed, Failed).

2. Motivation Public safety relies heavily on the roadworthiness of passenger vehicles. For fleet operators and the City of Chicago, inspection failures result in downtime, revenue loss, and administrative bottlenecks. By predicting whether a vehicle will pass or fail its inspection *before* it occurs, operators could proactively address maintenance issues, and the city could optimize inspector scheduling. This project aims to minimize operational inefficiencies and improve public safety compliance.

3. Method Data Preprocessing: We will clean the data by handling missing values, encoding categorical variables using One-Hot or Target Encoding, and normalizing numerical features. We will also engineer new features, such as "Vehicle Age" (Current Year - Model Year)

Classification (Supervised Learning): The primary task is binary classification to predict the inspection outcome (PASSED vs. FAILED). We will train and compare supervised models:

1. **Logistic Regression:** As a baseline for interpretability.
2. **Random Forest Classifier:** To handle non-linear relationships and feature interactions.
3. **Gradient Boosting (XGBoost or LightGBM):** For high performance on tabular data.
4. **Support Vector Machine (SVM):** To experiment with different kernels for separation.
5. **K-Nearest Neighbors (KNN):** To analyze similarity-based classification.

Unsupervised Analysis:

1. **Clustering (K-Means/DBSCAN):** We will cluster vehicles based on attributes like age, make, and inspection history to identify "risk groups".
2. **Anomaly Detection (Isolation Forest):** To identify unusual inspection patterns (e.g., brand new cars failing inspections or very old cars passing consistently) which might indicate data entry errors or fraud.

4. Intended Experiments

- **Performance Metrics:** We will evaluate models using Accuracy, Precision, Recall, F1-Score, and ROC-AUC. Given that "Failed" inspections might be the minority class, we will focus on Recall to ensure we catch potential failures.
- **Hyperparameter Tuning:** We will use Grid Search or Random Search to optimize parameters (e.g., tree depth for Random Forest, regularization for Logistic Regression).
- **Feature Importance Analysis:** We will determine which factors (e.g., Vehicle Age vs. Company Name) are the strongest predictors of inspection failure.